

## Paper 2 - Set 14 - Solutions

Generated practice paper: answer key and worked solutions.

### Answer key

| Question | Answer | Topic                                   |
|----------|--------|-----------------------------------------|
| 1        | B      | Quadratic functions                     |
| 2        | A      | Straight lines                          |
| 3        | C      | Indices and surds                       |
| 4        | F      | Probability                             |
| 5        | C      | Identifying errors in proofs            |
| 6        | C      | The logic of arguments                  |
| 7        | C      | The logic of arguments                  |
| 8        | A      | Chains of reasoning                     |
| 9        | E      | Exponentials and logarithms             |
| 10       | A      | Mathematical proof                      |
| 11       | B      | Mathematical proof                      |
| 12       | C      | Exponentials and logarithms             |
| 13       | E      | Triangles, radians and sectors          |
| 14       | E      | Chains of reasoning                     |
| 15       | F      | Differentiation                         |
| 16       | B      | The logic of arguments                  |
| 17       | G      | Differentiation                         |
| 18       | C      | Mathematical proof                      |
| 19       | H      | Simultaneous equations and inequalities |
| 20       | H      | Chains of reasoning                     |

## Worked solutions

### Question 1 QUA-02 · Quadratic functions · spec MM1.3

By completing the square, find the minimum value of  $2x^2 - 12x + 23$ .

**Answer: B**

$$2x^2 - 12x + 23 = 2(x - 3)^2 - 18 + 23 = 2(x - 3)^2 + 5, \text{ so the minimum is } 5 \text{ (at } x = 3\text{).}$$

### Question 2 LIN-06 · Straight lines · spec MM3.1

The perpendicular bisector of the line segment joining (1, 5) and (7, 1) has equation

**Answer: A**

Midpoint (4, 3); gradient of segment  $\frac{-4}{6} = -\frac{2}{3}$ , so the bisector has gradient  $\frac{3}{2}$ :  
 $y - 3 = \frac{3}{2}(x - 4) \Rightarrow 2y - 6 = 3x - 12 \Rightarrow 3x - 2y = 6.$

### Question 3 IND-02 · Indices and surds · spec MM1.1

Find the value of  $\left(\frac{27}{8}\right)^{-\frac{2}{3}}$ .

**Answer: C**

The negative index inverts the fraction:  $\left(\frac{8}{27}\right)^{2/3}$ . Cube root first:  $\left(\frac{2}{3}\right)^2 = \frac{4}{9}.$

### Question 4 PRO-03 · Probability · spec M7.7

$P(A) = 0.6$ ,  $P(B) = 0.5$  and  $P(A \text{ and } B) = 0.3$ . Which of the following statements is/are true?

- I  $A$  and  $B$  are independent.
- II  $A$  and  $B$  are mutually exclusive.
- III  $P(A \text{ or } B) = 0.8$

**Answer: F**

$P(A)P(B) = 0.3 = P(A \text{ and } B)$ , so independent (I). Not mutually exclusive since they can both happen (II false).  
 $P(A \text{ or } B) = 0.6 + 0.5 - 0.3 = 0.8$  (III).

### Question 5 ERR-14 · Identifying errors in proofs · spec Err2

Claim: if  $x$  and  $y$  are real with  $x^2 = y^2$ , then  $x = y$ .

Which of the following is a counterexample to the claim?

**Answer: C**

A counterexample must satisfy the hypothesis ( $x^2 = y^2$ ) and fail the conclusion ( $x \neq y$ ).

**Question 6** LGC-11 · The logic of arguments · spec Arg1

The statement 'if a triangle is equilateral then it is isosceles' is true. Which of the following must therefore also be true?

- I If a triangle is isosceles then it is equilateral.
- II If a triangle is not isosceles then it is not equilateral.
- III If a triangle is not equilateral then it is not isosceles.

**Answer: C**

Only the contrapositive (II) is logically equivalent to the original. I is the converse and III is the inverse; both are in fact false here.

**Question 7** LGC-02 · The logic of arguments · spec Arg2

Let  $P$  be the statement ' $x > 2$ ' and  $Q$  the statement ' $x^2 > 4$ ', where  $x$  is a real number. Which one of the following is true?

**Answer: C**

$x > 2 \Rightarrow x^2 > 4$ , so  $P$  is sufficient for  $Q$ . But  $x = -3$  gives  $x^2 > 4$  without  $x > 2$ , so  $P$  is not necessary.

**Question 8** RSN-10 · Chains of reasoning · spec Prf5

Each of 100 cards has a different integer from 1 to 100 written on it. What is the smallest number of cards that must be drawn (without looking) to be certain of having two cards whose numbers sum to 101?

**Answer: A**

The 50 pairs  $\{1, 100\}, \{2, 99\}, \dots, \{50, 51\}$  each sum to 101. Fifty cards (one from each pair) may avoid it; 51 cannot.

**Question 9** LOG-16 · Exponentials and logarithms · spec MM5.1

For which of the following values of  $x$  is  $3^x < 2^{x+1}$ ?

- I  $x = 0$
- II  $x = 1$
- III  $x = 2$

**Answer: E**

$x = 0$ :  $1 < 2$  true.  $x = 1$ :  $3 < 4$  true.  $x = 2$ :  $9 < 8$  false. So I and II.

**Question 10** PRF-20 · Mathematical proof · spec Prf3

$T_n$  denotes the  $n$ th triangular number  $\frac{n(n+1)}{2}$ . A student observes  $T_1 + T_2 = 4$ ,  $T_2 + T_3 = 9$ ,  $T_3 + T_4 = 16$  and conjectures that  $T_n + T_{n+1}$  is always a perfect square. Which of the following proves the conjecture?

**Answer: A**

Only A is a general algebraic verification.

**Question 11** PRF-17 · Mathematical proof · spec Prf1

Which of the following claims is FALSE, and therefore can be disproved by a counterexample?

**Answer: B**

$a = 1, b = -2$ :  $1 > -2$  but  $1 < 4$ . The others are true.

**Question 12** LOG-07 · Exponentials and logarithms · spec MM5.2

Which of the following statements is/are true for all positive real  $x$  and  $y$ ?

**I**  $\log_{10}(x + y) = \log_{10} x + \log_{10} y$

**II**  $\log_{10}(xy)^2 = 2\log_{10} x + 2\log_{10} y$

**III**  $\log_{10} x - \log_{10} y = \log_{10} x \div \log_{10} y$

**Answer: C**

II is correct:  $\log(xy)^2 = 2\log(xy) = 2\log x + 2\log y$ . I and III confuse the laws: log of a sum has no simple form, and a difference of logs is log of a quotient, not a quotient of logs.

**Question 13** TRI-12 · Triangles, radians and sectors · spec MM4.1

A triangle has sides  $a, b, c$  with  $a^2 + b^2 < c^2$ . Which of the following statements is/are true?

**I** The angle opposite  $c$  is obtuse.

**II**  $c$  is the longest side.

**III** The triangle cannot be isosceles.

**Answer: E**

Cosine rule:  $\cos C = \frac{a^2 + b^2 - c^2}{2ab} < 0$ , so  $C$  is obtuse (I). The largest angle is opposite the longest side, so  $c$  is longest (II). III is false:  $a = b = 1, c = 1.5$  works.

**Question 14** X2-26 · Chains of reasoning · spec Prf5

The positive integers  $a, b$  and  $c$  satisfy  $a < b < c$  and  $a + b + c = 12$ . Which of the following statements must be true?

**I**  $a \leq 3$

**II**  $c \geq 5$

**III**  $b = 4$

**Answer: E**

Since  $a < b < c$ ,  $3a < 12$  so  $a \leq 3$  (I);  $3c > 12$  so  $c \geq 5$  (II). III fails:  $1 + 2 + 9 = 12$  has  $b = 2$ .

**Question 15** X2-16 · Differentiation · spec MM6.3

The function  $f$  has  $f'(x) = (x - 1)(x - 3)^2$ . Which of the following statements is/are true?

**I**  $f$  has a local minimum at  $x = 1$ .

**II**  $f$  has a local maximum at  $x = 3$ .

**III**  $f$  is increasing for all  $x > 1$ .

**Answer: F**

$f'$  changes from negative to positive at  $x = 1$  (minimum). At  $x = 3$  the squared factor keeps  $f' \geq 0$  on both sides, so no turning point: II is false, and  $f' \geq 0$  for  $x > 1$  (zero only at 3) makes III true.

**Question 16** LGC-18 · The logic of arguments · spec Arg4

A statement reads: **\*\*for every positive integer  $n$ , there is a prime  $p$  with  $n < p < 2n$ .\*\*** Which of the following is its negation?

**Answer: B**

Negate the quantifiers in turn: 'for every  $n$ , there exists  $p$  with ...' becomes 'there exists  $n$  such that for every prime  $p$ , not ...', i.e. no prime lies strictly between  $n$  and  $2n$ .

**Question 17** DIF-19 · Differentiation · spec MM6.1

Which of the following statements is/are true for every function  $f$  that has a derivative at every real number?

- I If  $f'(a) = 0$  then  $f$  has a maximum or a minimum at  $x = a$ .
- II If  $f$  has a local minimum at  $x = a$  then  $f'(a) = 0$ .
- III If  $f'(x) > 0$  for all  $x$ , then  $f(3) > f(2)$ .

**Answer: G**

I is false:  $f(x) = x^3$  has  $f'(0) = 0$  but no turning point. II is true (a stationary point is required at an interior local extremum of a differentiable function). III is true: a strictly increasing function has  $f(3) > f(2)$ .

**Question 18** PRF-06 · Mathematical proof · spec Prf1

Which of the following statements can be disproved by a single counterexample?

- I Every prime number greater than 2 is odd.
- II Every integer of the form  $4k + 3$  is prime.
- III There exists an even prime number.

**Answer: C**

Only a 'for all' statement can be disproved by one counterexample. I is true (no counterexample exists). II is false and  $k = 3$  (15) disproves it. III is an existence statement - a counterexample is meaningless; it is proved by the example 2.

**Question 19** SIM-11 · Simultaneous equations and inequalities · spec MM1.5

Given that  $|2x - 3| < 5$ , which of the following statements is/are true?

- I  $-1 < x < 4$
- II  $x^2 < 16$
- III  $x > -1$

**Answer: H**

$-5 < 2x - 3 < 5 \Rightarrow -1 < x < 4$ , so I is true and III follows from I. For II: every  $x$  in  $(-1, 4)$  has  $x^2 < 16$  (the largest possible  $|x|$  is just under 4), so II is also true.

**Question 20** RSN-02 · Chains of reasoning · spec Prf5

The integers  $a, b, c$  satisfy  $a + b + c = 0$  and  $abc = 12$ . Which of the following statements must be true?

- I Exactly one of  $a, b, c$  is positive.
- II  $a^2 + b^2 + c^2$  is even.
- III One of  $a, b, c$  is divisible by 3.

**Answer: H**

$abc > 0$  means either none or exactly two of them are negative; they cannot all be positive with sum 0, so exactly two are negative and one positive (I).  $a^2 + b^2 + c^2 = (a + b + c)^2 - 2(ab + bc + ca) = -2(ab + bc + ca)$  is even (II). 3 is prime and divides  $abc$ , so it divides one of them (III). (Example:  $-1, -3, 4$ .)